# ***TechQuake* – Visualizing Layoffs in the Technological Landscape**

**Links: Crunchbase dataset // Yahoo Finance dataset**

## Introduction

Over the last three years, the country's technology industry has been severely impacted by workforce adjustments. To date, hundreds of thousands of tech workers have been laid off. Due to the unpredictability of this phenomenon, many are worried that they might be out of a job tomorrow. This concern extends to prospective workers who plan to work in the industry.

Hence, we were motivated to create an interface that is capable of providing insights for the U.S. workforce to learn the risk of layoffs in the industry from multiple angles, including temporal, geographical, subsector, and the involved companies. We also hope that this app can help inform economic planners and company executives about developing effective layoff mitigation strategies. As guidance, we focus on these questions of interest:

1. How have tech layoffs evolved over time? Can they be correlated with major economic or industry-specific events?
2. Which subsector within the tech industry is most susceptible to workforce adjustments?
3. Do the financing stages of a tech company impact layoffs?
4. How concentrated are the tech layoffs in the U.S.?

Upon using *TechQuake*, we promise that users will be more prepared to face this deepening crisis.

## Literature Review

In 2022, a total of 164,969 workers had been laid off and the number increased to 262,682 in 2023 (Helhoski, 2024). This is mainly due to the hiring spree after the pandemic. After a massive economic expansion in 2021, many companies find themselves having too many employees thus forcing them to lay off some workers for cost minimization (Chan, 2024). Furthermore, since many duties no longer require an employee to be physically present, businesses are opting to outsource some of their work. Hiring full-time employees costs them more, and outsourcing labour are much cheaper. For instance, Google intends to reorganize its financial team in Mexico and India (Elias, 2024).

Since this is a recent issue, most of the data visualization is from news articles. Past graphs mainly focus on the general trend of the layoffs. In an article by Perry (2022), the layoff trend was shown using bar plot and he notes that the reader will be able to perceive the growing scale of layoffs from this visualization. Therefore, bar plot is an excellent place to start when it comes to layoff visualizations since it is easy to grasp the overall trend. Furthermore, the website *Incredible Charts* gives an insight that layoffs might reflect a company's financial health (Twiggs, 2009). The simplest way to depict this is through plotting line plots for stock prices over time and adding a vertical dashed line representing the layoff dates to see how it correlates with the price movement. This allows us to unravel potential trends between layoff dates and the company's financial health.

It could also be useful to look at information about the top companies that are terminating employees. In her piece, Perry (2022) also included a bubble plot that displays the number of layoffs of the top corporations as users hover over the bubble. Encoding the bubbles' size to represent the magnitude of layoffs allows users to compare the scale of layoff between companies.

Next, we wish to examine firm attributes to discern whether there are any distinct layoff patterns. One appropriate strategy here is to make a comparative boxplot. The boxplot effectively summarizes the distribution of firm attributes related to layoffs, allowing for quick identification of variability and outliers (Suvarnapathaki, 2023).

By mixing general trends and focused visualizations in the application, the real challenge is to integrate all of them together. One way to do this is through linked brushing. In Kris' (2024) notes, the linked brushing between the histogram and the scatter plot allows a subset of data selected on one graph to be viewed on another graph. This interactive linking provides users with the capability to delve into specific segments of the data, thus facilitating more detailed and focused examinations of layoffs across different time periods. This concept eases the process of integrating all graphs together while maintaining the relationship between different graphs.

## Design

Recalling our Project Milestone 1 discussion, we decided to adopt the **Design Studio** modality for our project. This choice has allowed us to develop an interactive dashboard that presents the data intuitively for the users to engage and unpack unique tech layoff patterns.

We spent most of the project designing the individual components of the application. We first improved on Perry's idea by creating a dual-axis bar chart to facilitate monthly comparisons of tech layoffs in 2022-2024 and the companies involved. Apart from discerning potential temporal trends in layoffs, users can look for correlations between the volume of tech layoffs and its spread across companies. If both metrics peak simultaneously, it might suggest industry-wide issues, whereas discrepancies (high layoffs, few companies) might indicate problems specific to certain companies. For consistency, we color-coded the layoffs to be orange and the companies to be blue.

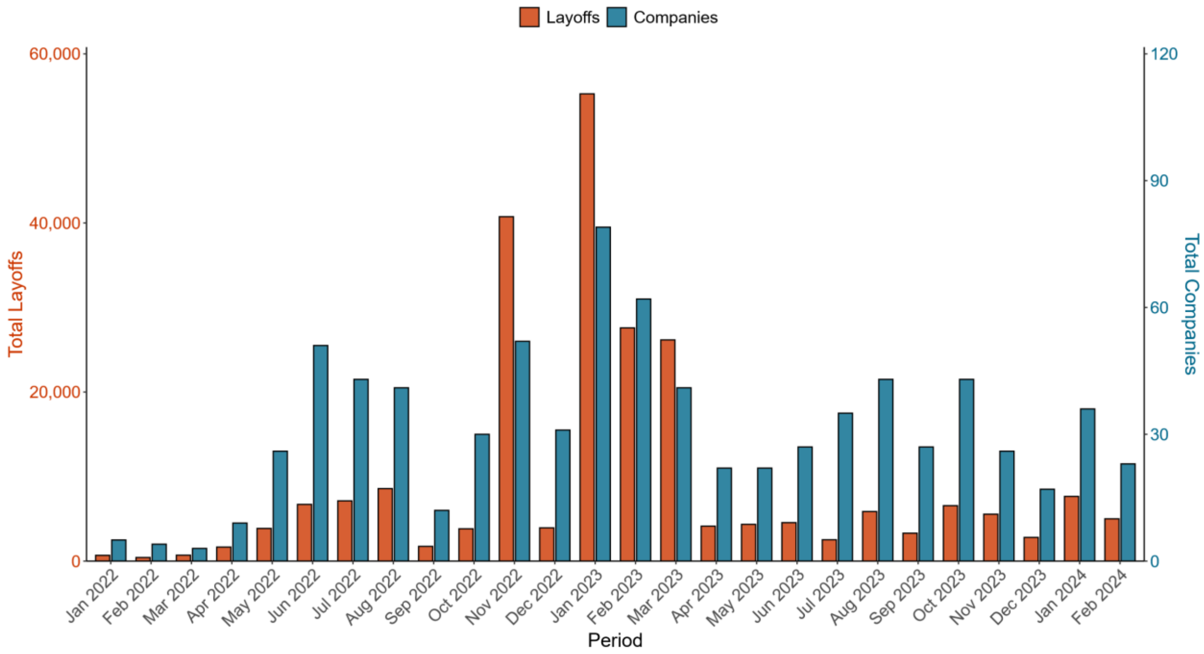



**Figure 1:** Tech Layoffs in 2022-2024.

In addition to the bar plot, we developed an interactive line graph of companies' stock prices. This specific focus assists stakeholders in understanding the layoff pattern with regards to a company's financial status at the time of its job cuts. Different phases of Covid-19 are integrated as rectangular boxes in the plot spanning the duration of each phase. To increase information density, we added vertical dashed lines which indicate the layoff dates of chosen companies from the dropdown menu. When comparing between two companies, we normalized each company's stock prices to show the troughs and peaks of smaller companies when compared to giant companies.

Next, we plotted a heatmap to identify the tech subsectors most prone to workforce adjustments. Our initial design was to use a log scale of total layoffs and arrange the subindustries by decreasing monthly layoffs. However, this approach was susceptible to misinterpretation as some sectors are significantly larger than the others. Thus, we decided to use mean layoffs as it illustrates the relative differences of layoffs between the subindustries of unequal sizes more effectively. Prospective tech employees can leverage this comprehensive heatmap to plan out which subindustry they should enter.

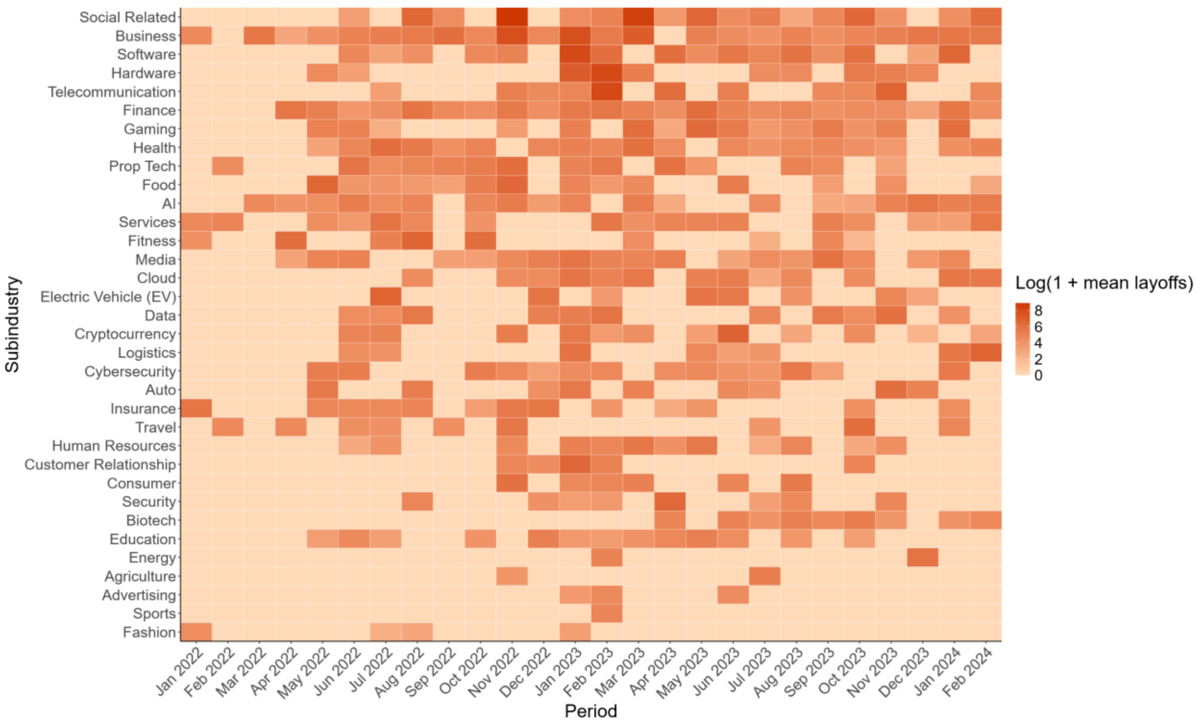
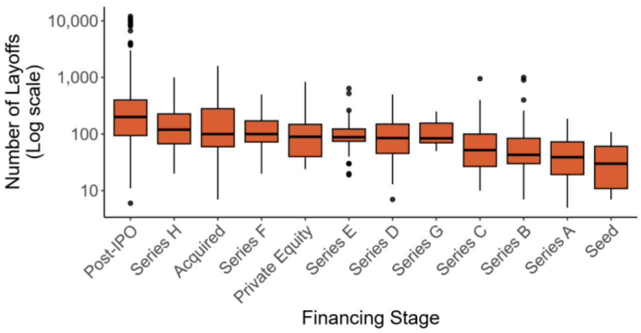



**Figure 2:** Tech Layoffs within Subindustries.

Then, we took inspiration from Suvarnapathaki and constructed side-by-side boxplots along with a summary table to investigate if layoffs vary by funding stages. We had to exclude several stages with insufficient observations to ensure that the layoff distributions are not misleading. To make comparisons between financing rounds easier, we ordered the boxplots by decreasing median number of layoffs. By analyzing the spread of layoffs across all financing rounds, users can deduce which company types would offer the best job stability. For economic planners, such insights could inform them about implementing strategic policies to help vulnerable companies.




| Financing Stage | Max | Q3 | Median | Mean | Q1 | Min |
|---|---|---|---|---|---|---|
| Post-IPO | 12000 | 400.00 | 200 | 618.14 | 94.00 | 6 |
| Series H | 1000 | 230.00 | 121 | 219.29 | 69.25 | 20 |
| Acquired | 1600 | 281.25 | 100 | 217.36 | 59.75 | 7 |
| Series F | 500 | 171.25 | 100 | 137.18 | 72.50 | 20 |
| Private Equity | 840 | 148.50 | 90 | 187.36 | 40.00 | 24 |
| Series E | 640 | 123.00 | 88 | 121.30 | 74.50 | 19 |
| Series D | 500 | 150.00 | 85 | 108.66 | 45.50 | 7 |
| Series G | 250 | 156.00 | 84 | 123.44 | 70.00 | 50 |
| Series C | 950 | 100.00 | 52 | 83.66 | 26.75 | 10 |
| Series B | 1000 | 84.25 | 43 | 88.26 | 30.00 | 7 |
| Series A | 187 | 72.75 | 39 | 53.73 | 19.25 | 5 |
| Seed | 109 | 68.50 | 30 | 43.86 | 12.00 | 7 |

**Figure 3:** Tech Layoffs by Financing Stage.

Additionally, we chose to use two pie charts to reveal patterns in layoff concentration in the U.S. The static pie chart shows the overall layoff landscape of U.S. cities. Cities with smaller values of layoff percentage are combined together to improve visibility of the design. Furthermore, hovering over the chart displays the user the number of layoffs happening in that city. This allows them to learn the risk of working with companies located in certain cities. The second pie chart serves as a comparison tool for users to compare layoff numbers between specific cities chosen within the dropdown. This provides additional insight to the first chart where job seekers can compare how vulnerable it is to work in a certain city.

On top of these visualizations, we implemented Perry's bubble plot concept of companies with most layoffs. We figured that audiences might be interested in identifying the biggest culprits of job cuts. This graph not only highlights the scale of layoffs at each company but also provides a clear visual comparison among them. Users can set a cutoff to display a specific number of tech companies. They can also hover over each bubble to obtain more information about the company.

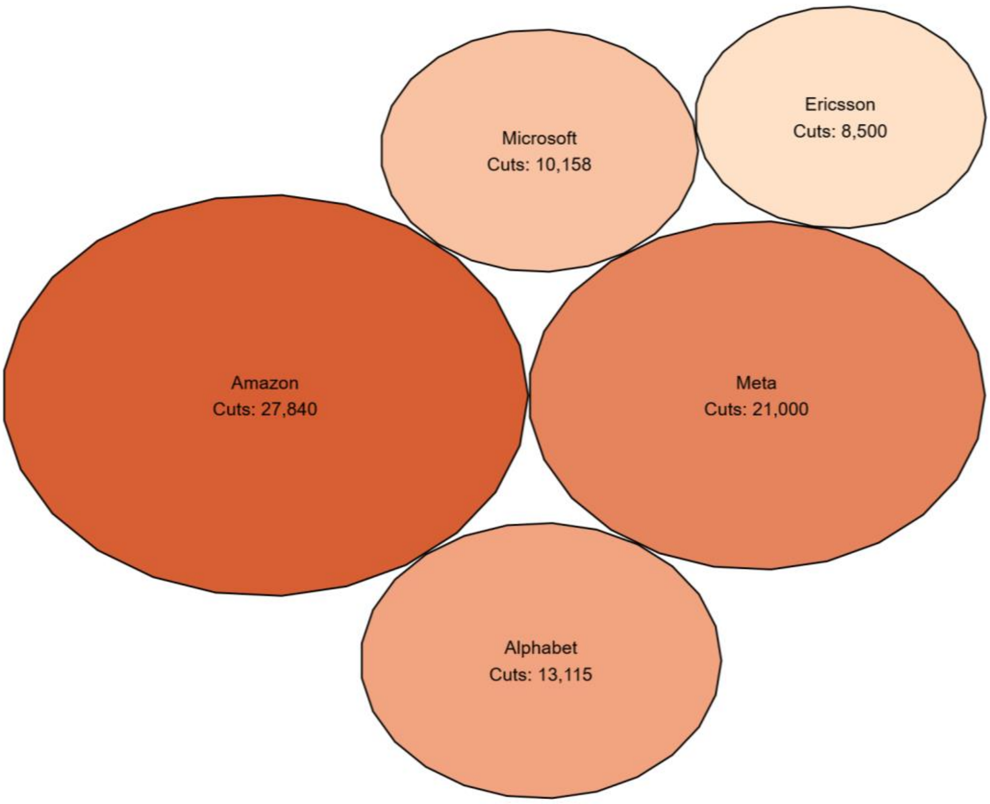
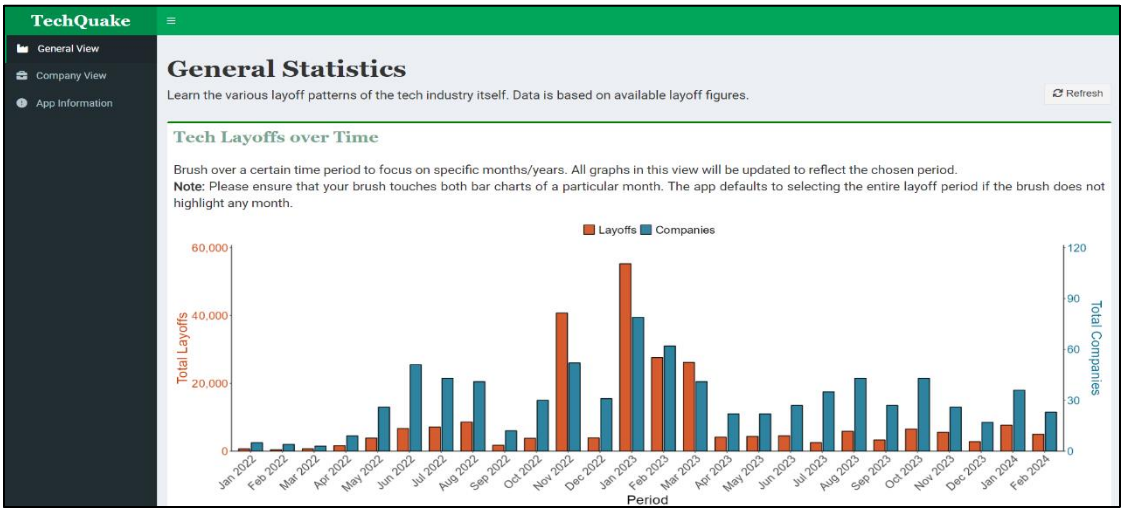


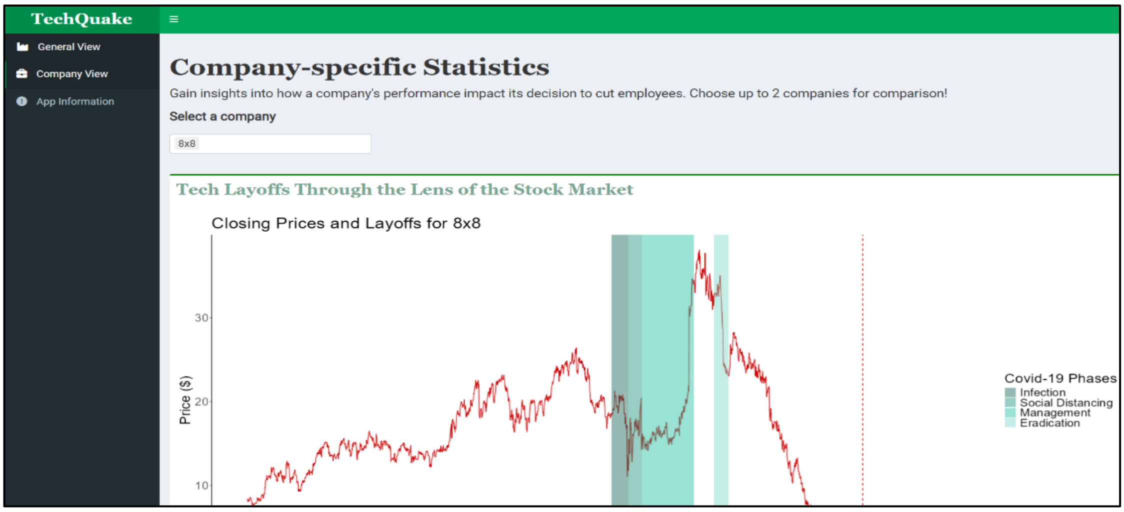

**Figure 4:** Companies with Most Layoffs.

With these design choices finalized, we integrated them into a unified dashboard application. The interface features multiple tabs: a general overview of tech layoffs, a company-specific view, and an information section. We dedicated a specific tab for the company's market performance plot to give space for the detailed information and to avoid cluttering the general tab.

On the other hand, the general tab houses our remaining visualizations. We were mindful of the possibility that users could feel overwhelmed by the information density of these graphs. Therefore, we limited the page to show no more than three different trends at a time but gave users the option to customize the type of visualization displayed for the third graph. Additionally, we chose to display the temporal trend of tech layoffs and the companies with most job cuts as the first two graphs because these visualizations provide the fundamental understanding into this deepening crisis.

Whilst designing this tab, we also realized that there is additional value in being able to analyze specific intervals rather than only examining the entire tech layoff period. For instance, geographical trends in tech layoffs from 2022 might have differed significantly from those observed last year. There might also have been technological changes that influenced a period of layoff within specific subsectors. To accommodate these hidden insights, we implemented dynamic linking across all the general plots. Users can brush over the dual-axis bar chart to select a certain timeframe and the other graphs will be updated to reflect this layoff period. We are confident that this feature will elevate the users' understanding of this phenomenon if it is used properly.






**Figures 5 & 6:** Snippets of the *TechQuake* dashboard.

## <u>Conclusion</u>

Our project is capable of shedding light on layoff trends in the tech industry. Navigating through the app allows stakeholders to reveal differences in layoff magnitude in various sub industries and financing rounds. Economic planners can use this information to introduce targeted policies for most vulnerable companies. Job seekers can make more informed decisions before entering the labor market. Companies' executives can plan their courses informatively and will be able to create better resilience as preparation for contrasting conditions.

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